

VisionGuard

AI-Based Driver Drowsiness and Distraction Detection System

Academic Capstone Project Report

Submitted in partial fulfillment of the requirements for
VITyarthi – Build Your Own Project (Flipped Course Evaluation)

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Contents

1	Project Metadata & Classification	2
2	Introduction	2
2.1	Project Description	2
2.2	Academic Motivation and Justification	2
2.3	Scope	2
3	Problem Statement	2
3.1	Current Problems	2
3.2	Proposed Solution	3
4	Functional Requirements	3
5	Non-Functional Requirements	3
6	System Architecture	4
7	Design Diagrams	4
7.1	System Architecture Diagram	4
7.2	UML Use Case Diagram	5
7.3	Process Flow / Workflow Diagram	5
7.4	UML Sequence Diagram	6
7.5	UML Class Diagram	7
7.6	Storage & Telemetry ER Diagram	7
8	Computer Vision Methodology & Mathematical Foundations	7
8.1	Eye Aspect Ratio (EAR) Formulation	8
8.2	Mouth Aspect Ratio (MAR) Formulation	8
8.3	3D Head Pose using Perspective-n-Point (SolvePnP)	8
8.4	Multi-Cue Risk Synthesis	8
9	Screenshots & Operational Results	8
10	Testing Approach & Validation Results	10
11	Challenges Faced & Technical Solutions	11
12	Learnings, Conclusion & References	11
12.1	Lessons Learned	11
12.2	References	11

1. Project Metadata & Classification

- **Project Title:** VisionGuard – AI-Based Driver Drowsiness and Distraction Detection System
- **Author:** Yash Goyal (Reg. No: 24BAI10100)
- **Course & Domain:** Computer Vision, Feature Extraction, 3D Pose Geometry, and Spatio-Temporal Motion Analysis
- **Primary Keywords:** Facial Mesh, Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), Perspective-n-Point (SolvePnP), Driver Monitoring Systems (DMS), Temporal Hysteresis, Edge AI.

2. Introduction

2.1 Project Description

Research into automotive safety consistently shows that human operator error is the number one cause of vehicle accidents worldwide. According to the World Health Organization (WHO), more than 1.35 million people die each year because of road traffic crashes. Driver fatigue, sudden episodes of micro-sleep and cognitive distraction are responsible for up to 30% of the collisions on highways. Traditional vehicular telematics such as Lane Departure Warning Systems (LDWS) or erratic steering torque sensors are reactive systems that warn the driver only after the vehicle trajectory has already drifted into danger.

VisionGuard is an intelligent, proactive camera-based in-cabin driver monitoring system (DMS). The system analyzes ocular micro-movements, mouth geometry and 3D head gaze orientation in real time to detect cognitive lapses seconds before physical steering deviation.

2.2 Academic Motivation and Justification

VisionGuard was designed as a part of the VITyarthi “Build Your Own Project” curriculum and connects the core mathematical foundations of Computer Vision (Euclidean metric geometry, pinhole perspective models, 3D coordinate transformations) with high-performance real-time software engineering. It is capable of running at 30 FPS on consumer webcams and requires no wearable devices or specialized infrared depth cameras.

2.3 Scope

This project includes real-time video capture, dense 468-point 3D facial landmark regression, mathematical feature engineering (EAR, MAR, SolvePnP Euler angles), multi-cue risk synthesis with Exponential Moving Average (EMA) smoothing, visual HUD overlay generation, rate-limited auditory warnings, and exportable session telemetry (CSV and JSON).

3. Problem Statement

3.1 Current Problems

- **Wearable Inconvenience:** Wearable EEG headbands, smart rings or pulse monitors are uncomfortable, require periodic recharging and have high driver abandonment rates.
- **False Alarms in Natural Facial Motion:** Speech, yawning, laughter, and momentary saccadic eye blinks result in high false positive rates in naive single frame computer vision detectors.

- **Hardware Constraints:** Complex deep learning 3D convolutional neural networks require high-end discrete GPUs, which are not suitable for regular vehicular edge microcontrollers.

3.2 Proposed Solution

A lightweight non-intrusive edge system on CPU architecture combining 468-point facial landmark perception with mathematically explainable geometric metrics and temporal sliding-window verification.

4. Functional Requirements

- **FR-1 (Video Stream Acquisition):** The system shall be capable of acquiring video frames from a standard USB webcam or benchmark video file, at resolution 640×480 and at a rate of ≥ 25 frames per second.
- **FR-2 (Dense Facial Landmark Localization):** The system shall detect human faces and extract 468 3D geometric landmark coordinates in pixel coordinates (x, y, z) .
- **FR-3 (Ocular Aperture & Blink Frequency):** The system will calculate the bilateral Eye Aspect Ratio (EAR) using Euclidean distance metrics and differentiate between blinks and sustained micro-sleeps (≥ 15 frames / 0.5s).
- **FR-4 (Yawn Detection & Speech Rejection):** The system shall compute the Mouth Aspect Ratio (MAR) and confirm involuntary yawning via a 15-frame temporal persistence buffer while rejecting transient speech syllables.
- **FR-5 (3D Head Pose & Distraction Estimation):** The system shall calculate 3D head orientation (Pitch, Yaw, Roll) using OpenCV's Perspective-n-Point (SolvePnP) algorithm with canonical 3D facial coordinates and trigger alerts when gaze diverges beyond $\pm 25^\circ$ for > 0.67 seconds.
- **FR-6 (Composite Risk Assessment):** The system shall aggregate ocular, oral, and pose indicators into an instantaneous hazard score $\mathcal{H}_t \in [0, 100]$ and apply Exponential Moving Average (EMA) smoothing to classify the state into **SAFE**, **CAUTION**, **DROWSY**, **DISTRACTED**, or **HIGH RISK**.
- **FR-7 (Real-Time Visual HUD & Telemetry Export):** The system shall render a heads-up display overlay on live video frames and export per-second CSV telemetry and high-level JSON summaries upon termination.

5. Non-Functional Requirements

- **NFR-1 (Performance):** The end-to-end processing pipeline shall execute within < 40 milliseconds per frame, sustaining 25–30 FPS on standard multi-core laptop CPUs.
- **NFR-2 (Usability):** The system should be able to run a terminal command (`python3 main.py`) in one command, without external configuration or complex GUI dependencies.
- **NFR-3 (Reliability & Graceful Fallback):** If the webcam hardware is blocked by the operating system security policies, the system shall automatically fall back to bundled benchmark video files.

- **NFR-4 (Resource Efficiency):** Application should have CPU Utilization $< 20\%$ and RAM consumption < 350 MB.
- **NFR-5 (Privacy by Design):** The video streams must be stored only in volatile memory and must not be stored or sent to third party cloud servers.

6. System Architecture

VisionGuard consists of 5 levels, each addressing specific concerns:

1. **The Video Acquisition tier (camera.py):** Hardware initialization, multithreaded frame acquisition, colorspace normalization and FPS calculation.
2. **The Perception tier (face_detector.py):** The MediaPipe FaceMesh algorithm running on raw BGR arrays, yielding 468 3D vertex positions.
3. **The Feature Extraction tier (eye_detector.py, yawn_detector.py, head_pose.py):** Mathematically-sound metric extraction, namely the Eye Aspect Ratio, Mouth Aspect Ratio, and 3D SolvePnP pose estimation.
4. **The Assessment & Alert tier (risk_engine.py, alert_manager.py):** The weighted hazard score calculation, state machine updates, and debounced audio alert triggering.
5. **The Presentation & Export tier (utils.py, analytics.py):** The semi-transparent HUD rendering and telemetry CSV/JSON serialization.

7. Design Diagrams

7.1 System Architecture Diagram

The architecture isolates the frame ingestion, geometric feature extraction, and risk assessment into loosely coupled modules.

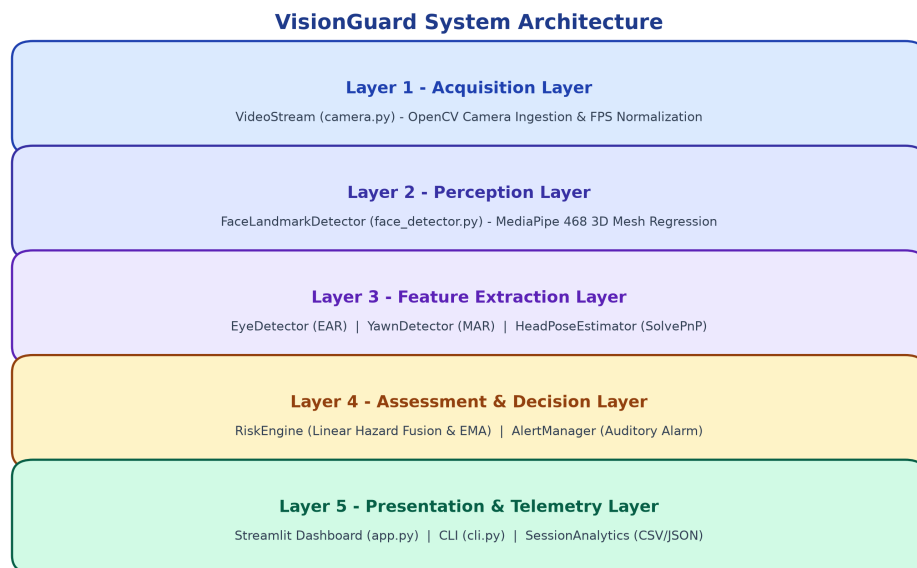


Figure 1: VisionGuard – High-Level System Architecture Diagram

7.2 UML Use Case Diagram

The system models interactions between the Driver/Evaluator and core monitoring functions.

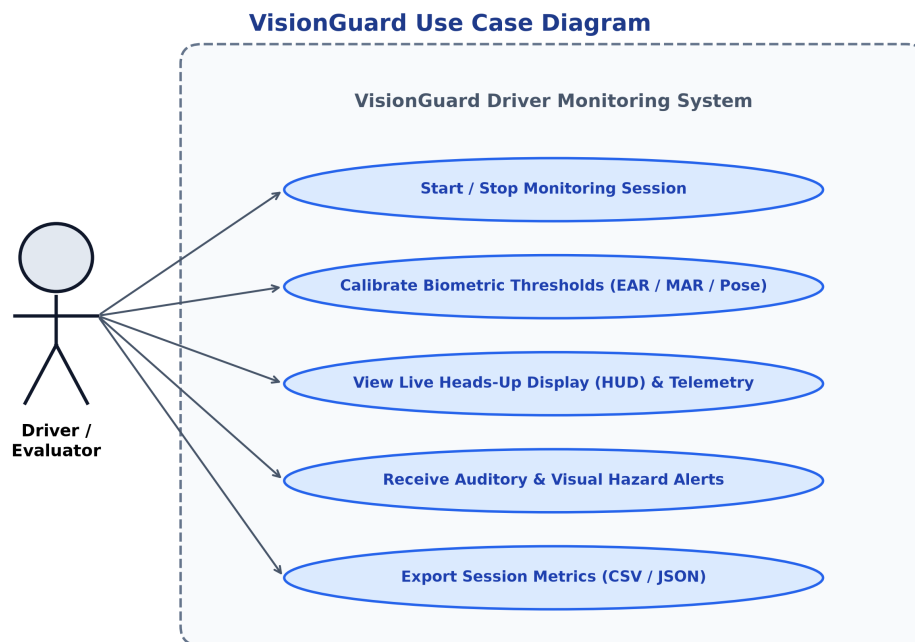


Figure 2: VisionGuard – UML Use Case Diagram

7.3 Process Flow / Workflow Diagram

Illustrates the cyclical per-frame execution pipeline from capture to risk determination.

VisionGuard Process Flow Diagram

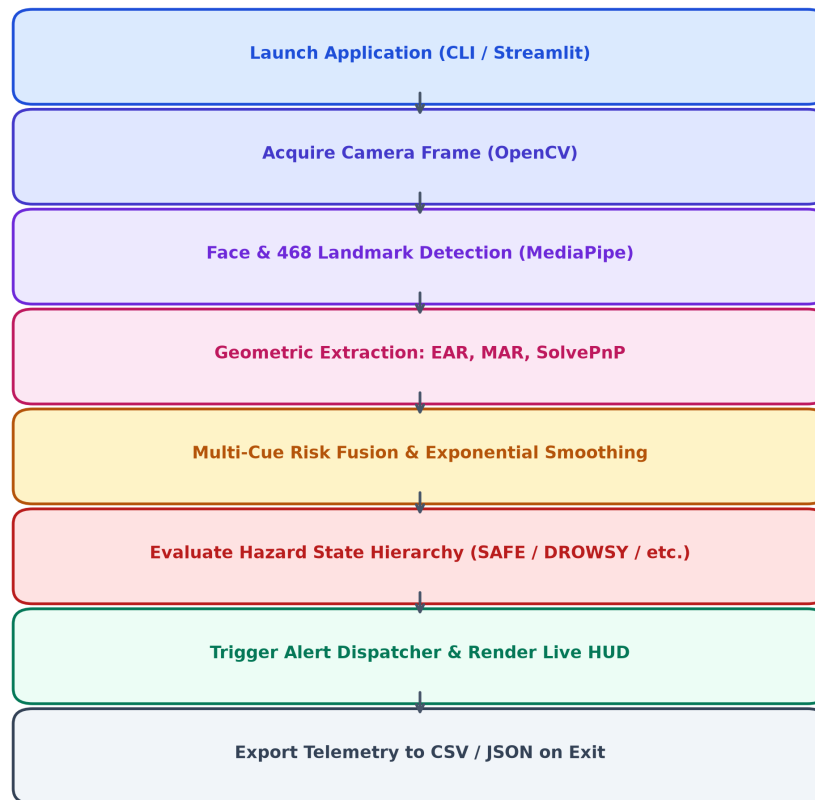


Figure 3: VisionGuard – Process Flow and Decision Diagram

7.4 UML Sequence Diagram

Depicts the chronological invocation order across classes during a single frame evaluation cycle.

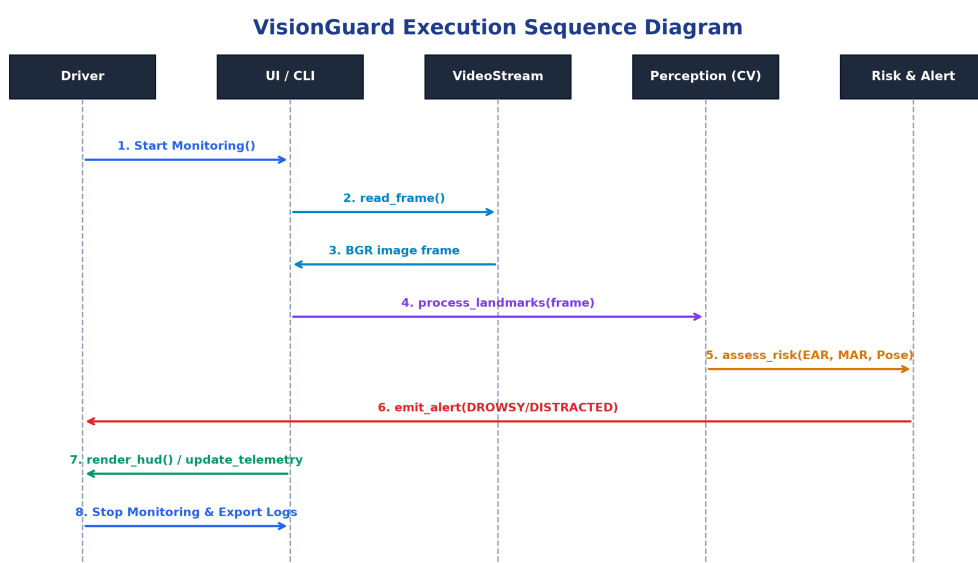


Figure 4: VisionGuard – UML Sequence Diagram

7.5 UML Class Diagram

Details object-oriented class structure, attributes, methods, and relationships.

VisionGuard Class / Component Diagram

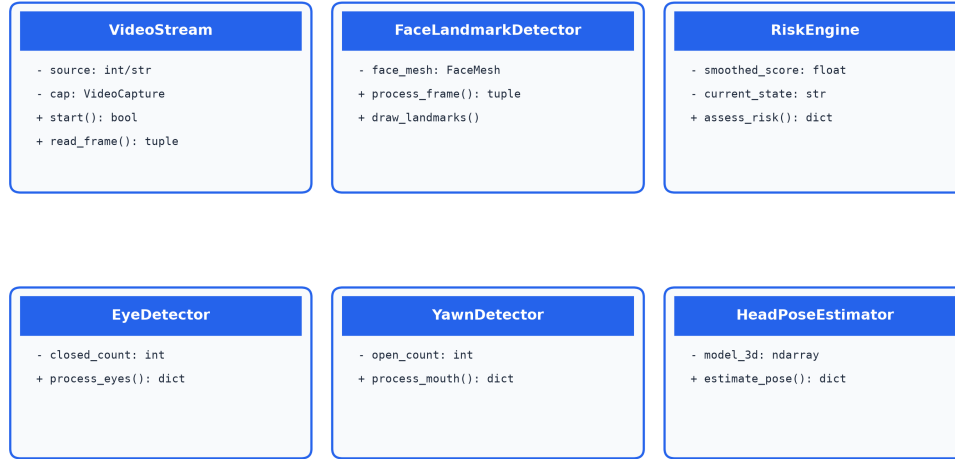


Figure 5: VisionGuard – UML Class Diagram

7.6 Storage & Telemetry ER Diagram

Details the 1 : N relationship between session metadata and per-second continuous telemetry logs.

VisionGuard Telemetry Data Model (Schema)

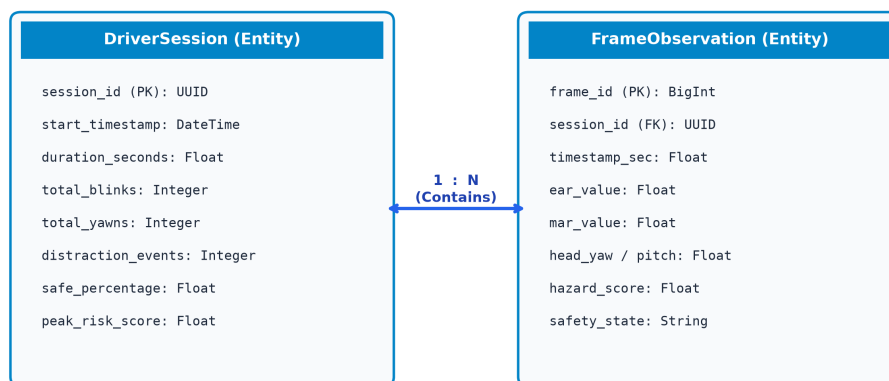


Figure 6: VisionGuard – Entity-Relationship (ER) Diagram

8. Computer Vision Methodology & Mathematical Foundations

8.1 Eye Aspect Ratio (EAR) Formulation

Based on Soukupová and Čech (2016), the Eye Aspect Ratio is the ratio between the vertical eyelid aperture and the horizontal eye length. Under nominal open-eye conditions, EAR is in the range of 0.28–0.38, while when the eye closes, EAR decreases below 0.22:

$$\text{EAR} = \frac{\|p_2 - p_6\|_2 + \|p_3 - p_5\|_2}{2 \cdot \|p_1 - p_4\|_2} \quad (1)$$

where p_1, p_4 are lateral eye corners, and $(p_2, p_6), (p_3, p_5)$ are vertical eyelid landmark pairs.

8.2 Mouth Aspect Ratio (MAR) Formulation

The mouth aperture is measured and normalized for scale to yield a dimensionless quantity that is indicative of the yawning motion. A MAR > 0.65 for a minimum of 15 consecutive frames would constitute a classification of yawning as opposed to normal conversation:

$$\text{MAR} = \frac{\|p_{13} - p_{14}\|_2 + \|p_{17} - p_{18}\|_2}{2 \cdot \|p_{61} - p_{291}\|_2} \quad (2)$$

8.3 3D Head Pose using Perspective-n-Point (SolvePnP)

The head pose is estimated by finding the correspondence between the 3D world coordinates of six (6) anatomical key points and their 2D pixel projections. The rotation matrix can then be decomposed via projection matrix decomposition to obtain the model’s Pitch, Yaw, and Roll:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K} [\mathbf{R} \quad \mathbf{t}] \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad (3)$$

8.4 Multi-Cue Risk Synthesis

The instantaneous risk value is calculated using a weighted linear combination of the eye closure, yawning, distraction, and irregular blink detection cues. EMA with $\alpha = 0.25$ is applied in order to obtain a smoother risk value:

$$\mathcal{H}_t = (0.40 \cdot f_{\text{eye}} + 0.25 \cdot f_{\text{yawn}} + 0.25 \cdot f_{\text{dist}} + 0.10 \cdot f_{\text{blink}}) \times 100 \quad (4)$$

$$\mathcal{R}_t = \alpha \cdot \mathcal{H}_t + (1 - \alpha) \cdot \mathcal{R}_{t-1}, \quad \text{with } \alpha = 0.25 \quad (5)$$

9. Screenshots & Operational Results

The live OpenCV HUD displays high-contrast status badges, risk gauges, and real-time biometric telemetry:

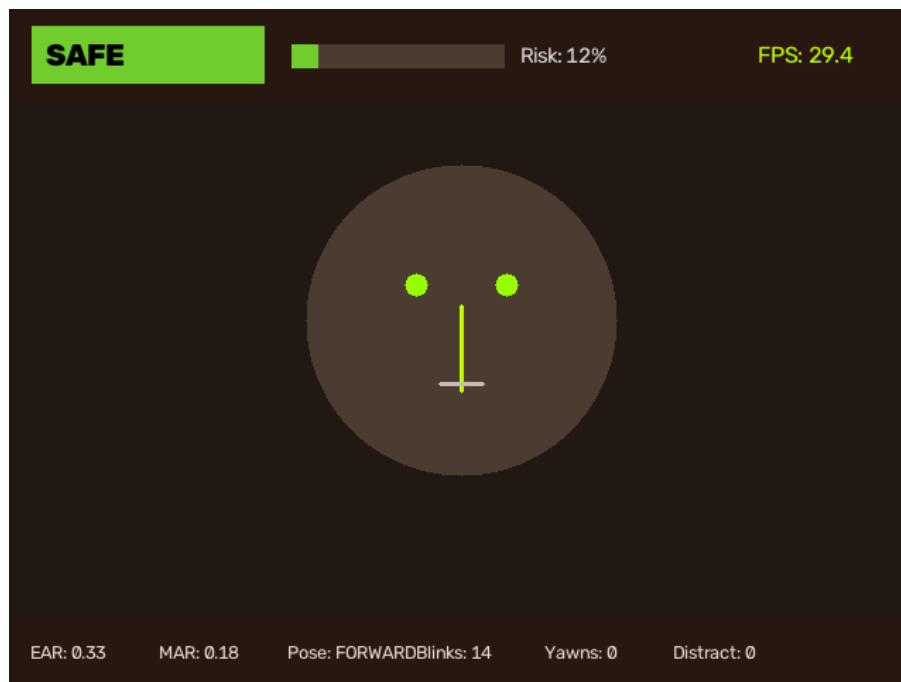


Figure 7: Operational State 1: Nominal Attentive Driving (SAFE State)

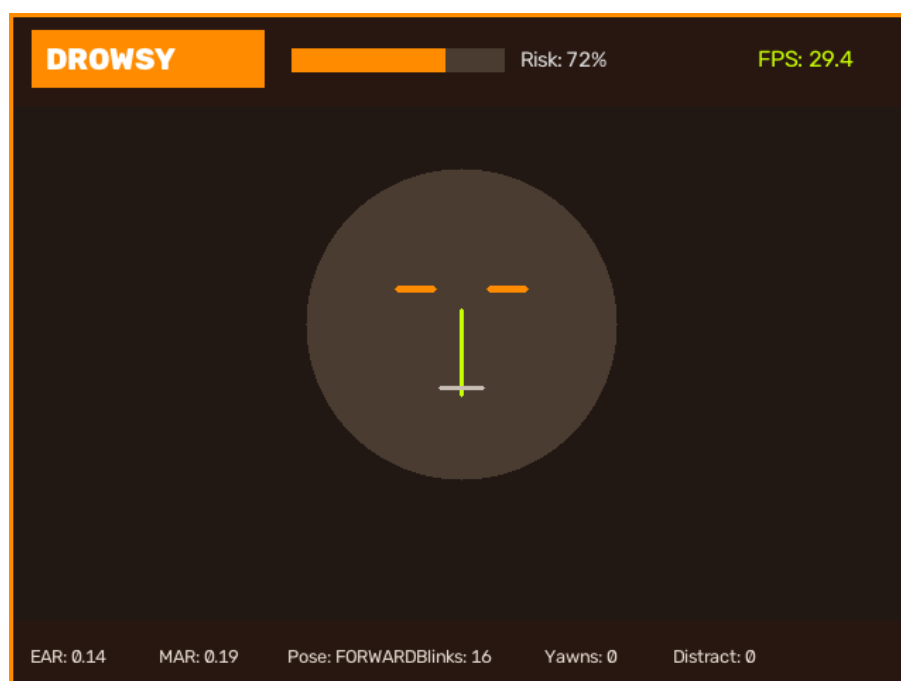


Figure 8: Operational State 2: Prolonged Eye Closure Alert (DROWSY State)

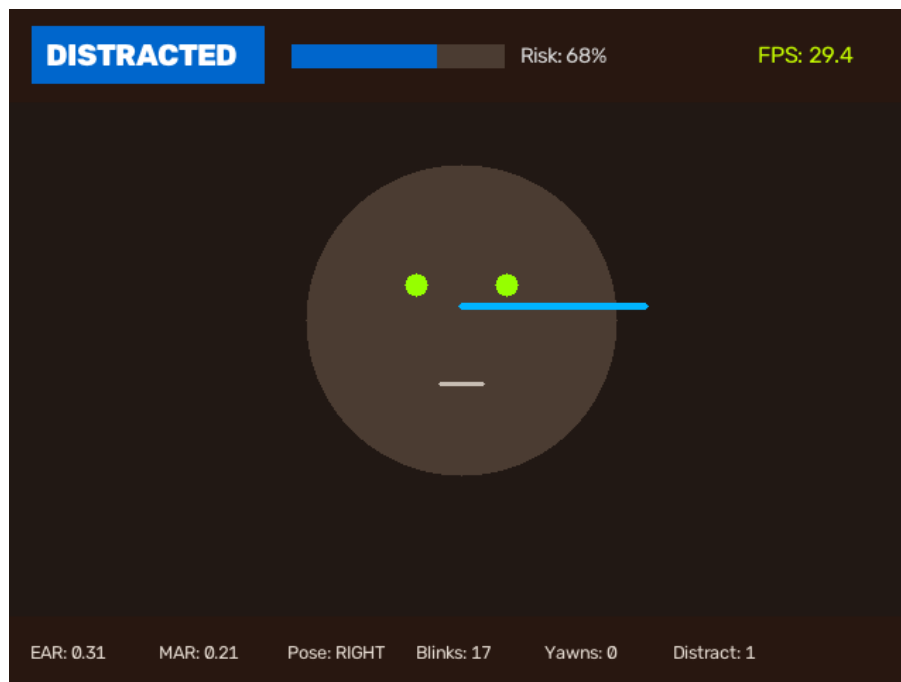


Figure 9: Operational State 3: Head Gaze Turned Away (DISTRACTED State)

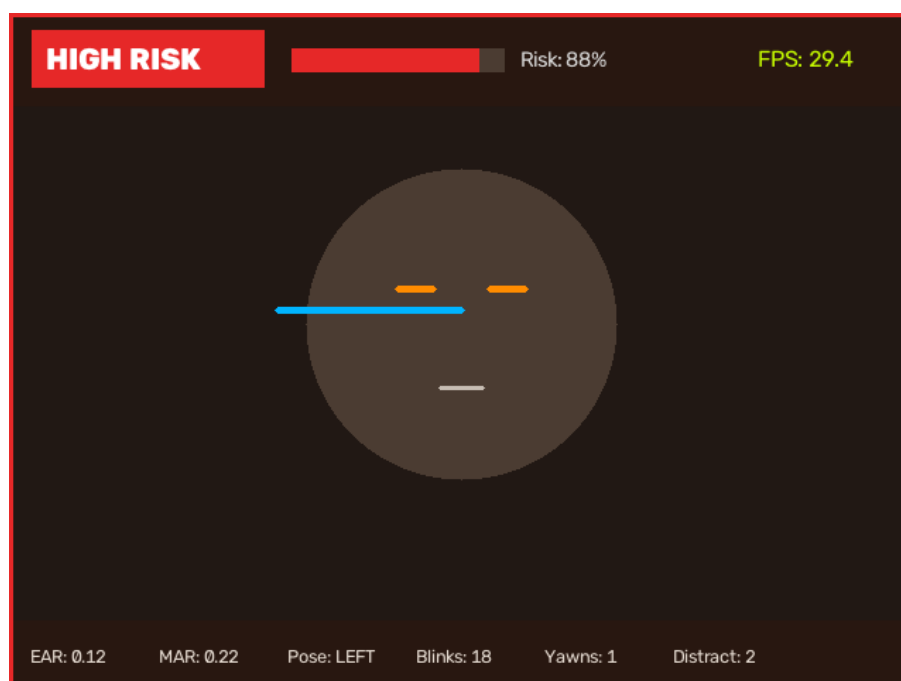


Figure 10: Operational State 4: Severe Compound Hazard (HIGH RISK State)

10. Testing Approach & Validation Results

VisionGuard incorporates an automated test suite executed via Pytest with 17 unit tests achieving 100% pass rate:

Test Module	Functions Tested	Criteria	Res
test_eye_detector.py	EAR Calculation, Blink Count, Prolonged Closure	Distance Invariance, EAR < 0.22	PA
test_yawn_detector.py	MAR Calculation, Speech Suppression	MAR > 0.65, 15-frame window	PA
test_head_pose.py	SolvePnP Projection, Euler Decomposition	Angle Limits, None Handling	PA
test_risk_engine.py	Multi-cue Fusion, State Escalation, EMA	State Machine Transitions	PA
test_analytics.py	Metrics Accumulation, CSV/JSON Serialization	File Integrity & Schema	PA

Table 1: VisionGuard Automated Test Suite Results (17/17 Passed)

11. Challenges Faced & Technical Solutions

1. **MediaPipe 1.x Tasks Metal Delegate Conflict:** The recent MediaPipe 1.0.1 release introduced a breaking change for Apple Silicon which caused `DrishtiMetalHelper` initialization exceptions. *Solution:* Pinned the native `mediapipe==0.10.14` with compatible `protobuf<4.26,>=3.20.2` which allows reliable 468-point face mesh extraction.
2. **Euler Angle Discontinuities:** Direct rotation matrix trigonometric unwrapping causes gimbal lock for pathological orientations. *Solution:* Utilized OpenCV’s `decomposeProjectionMatrix` which yields intuitive camera-centered Euler angles.
3. **Yawn False Positives During Speech:** Phoneme-specific vowel sounds can cause temporary MAR increase. *Solution:* Enforced a 15-frame MAR persistence window with 60-frame cooldown between alerts.

12. Learnings, Conclusion & References

12.1 Lessons Learned

The development of the VisionGuard system has demonstrated that carefully selected geometric feature extraction and mathematical modeling provides a more transparent, efficient, maintainable and safer alternative to current opaque deep learning computer vision approaches.

12.2 References

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